

STAT452

Variable Selection

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Why Perform Variable Selection?

Goals of Variable Selection:

- **Simplicity:** Remove redundant predictors.
Occam's Razor: The simplest adequate model is best.
- **Avoid Noise:** Unnecessary predictors add noise and obscure true relationships.
⇒ Wastes degrees of freedom and reduces precision.
- **Reduce Collinearity:** Too many variables may try to explain the same effect.
⇒ Leads to unstable coefficient estimates.
- **Minimize Cost:** Fewer predictors mean lower data collection costs and faster models.
⇒ Especially important in predictive modeling.

Preparation Before Selection:

- **Screen for Outliers and Influential Points**
⇒ Consider removing or treating them temporarily.
- **Apply Appropriate Transformations**
⇒ Normalize, log-transform, or standardize variables as needed.

The Temptation of Using p -Values

- In linear models, it's tempting to select variables based on their p -values.
- Idea: keep variables with significant p -values (small), drop those with large ones.
- Common practice among regression users – even statisticians.

Test Statistic and Hypothesis: we test $H_0 : \beta_i = 0$ using the (Wald) test statistic:

$$\frac{\hat{\beta}_i}{\text{se}(\hat{\beta}_i)}$$

Under H_0 , this has a $t_{n-(p+1)}$ distribution, and tends to standard normal as $n \rightarrow \infty$.

Why This is a Bad Idea

Recall the form of the test statistic:

$$\frac{\hat{\beta}_i}{\frac{\hat{\sigma}}{\sqrt{n \widehat{\text{Var}}(X_i)}} \sqrt{\text{VIF}_i}}$$

The denominator depends on:

- ① **Noise level** ($\hat{\sigma}$): More noise inflates the standard error $\Rightarrow$ lower significance.
- ② **Sample size** (n): Larger n shrinks the standard error $\Rightarrow$ higher significance.
- ③ **Predictor variance** ($\widehat{\text{Var}}(X_i)$): More spread in X_i improves estimation $\Rightarrow$ higher significance.
- ④ **Collinearity** (Variance Inflation Factor, VIF_i): High multicollinearity inflates the standard error $\Rightarrow$ lower significance.

Implication: The test statistic is sensitive to data quality and predictor structure—not just effect size!

What Affects Significance?

Factors that influence the size of the test statistic:

- ① **Larger coefficients ($\hat{\beta}_i$):**
⇒ Increases the numerator ⇒ higher test statistic ⇒ more significant.
- ② **Lower noise in the data ($\hat{\sigma}$):**
⇒ Reduces standard error in the denominator ⇒ increases test statistic.
- ③ **Larger sample size (n):**
⇒ Shrinks standard error ($\sim 1/\sqrt{n}$) ⇒ boosts significance even for small effects.
- ④ **Higher variance of predictor ($\text{Var}(X_i)$):**
⇒ Improves estimation precision ⇒ increases test statistic.
- ⑤ **Higher multicollinearity (measured by VIF_i):**
⇒ Inflates standard error ⇒ lowers test statistic ⇒ reduces significance.

What's the Real Problem?

- **The test statistic (and p-value) combines:**
 - The *size* of the coefficient, and
 - The *precision* of its estimate
- **This leads to misleading conclusions:**
 - Unimportant variables with high variance or low multicollinearity may appear significant
 - Important variables may appear non-significant if measured with high uncertainty
 - As $n \rightarrow \infty$, even tiny nonzero coefficients will become statistically significant
- **Variable selection based on p-values tends to favor:**
 - Predictors with high variance
 - Predictors with low correlation to others (low VIF)
 - Larger sample sizes, regardless of practical relevance

Implication: Selecting variables solely based on p -values can harm model interpretability and prediction.

Better alternatives:

- Cross-validation (CV)
- Information criteria (AIC, BIC)
- Regularization methods (LASSO, Ridge)

Goal of Variable Selection

- Choose a subset of predictors that results in a model that:
 - Predicts well on new data
 - Is interpretable and not overfit
- Using p -values alone is not reliable (as discussed)
- Better methods: use prediction-based metrics and data-driven selection

Cross-Validation for Variable Selection

- Cross-validation estimates the model's ability to predict **unseen** data.
- C_p , AIC, and BIC attempt to do this analytically; CV does it empirically.
- Two common forms:
 - **Leave-One-Out CV (LOOCV)**: low bias, higher variance, consistent but slow
 - **k -Fold CV** (typically $k = 5$ or 10): faster, more stable, often used in practice

Comparison: LOOCV vs. k -Fold CV

- **LOOCV:**

- Consistent but tends to overfit (includes unnecessary variables)
- More computationally expensive

- **k -Fold CV:**

- More efficient (only k models to train)
- Less variance in error estimates
- Slightly less accurate if models are all misspecified

Stepwise Variable Selection

- Adds or removes one variable at a time based on a selection criterion:
 - AIC, BIC, Adjusted R^2 , Mallows' C_p , CV error, etc.
- Types:
 - **Forward stepwise:** Start small, add one variable at a time
 - **Backward stepwise:** Start big, remove one variable at a time
 - **Mixed (forward-backward):** Add or remove based on best step

Akaike Information Criterion (AIC)

Definition:

$$\text{AIC} = -2 \log \hat{L} + 2p$$

- $\hat{L}$: maximum likelihood of the fitted model
- p : number of parameters (including intercept)

Interpretation:

- AIC estimates the *relative out-of-sample prediction error*.
- It penalizes model complexity to avoid overfitting.
- **Lower AIC** values indicate a better trade-off between fit and simplicity.

Use AIC when:

- Your goal is **prediction**, not necessarily finding the “true” model.
- You are comparing multiple models fit to the *same dataset*.

Bayesian Information Criterion (BIC)

Formula:

$$\text{BIC} = -2 \log \hat{L} + p \log n$$

- $\hat{L}$: Maximum likelihood of the fitted model
- p : Number of estimated parameters
- n : Sample size

Interpretation:

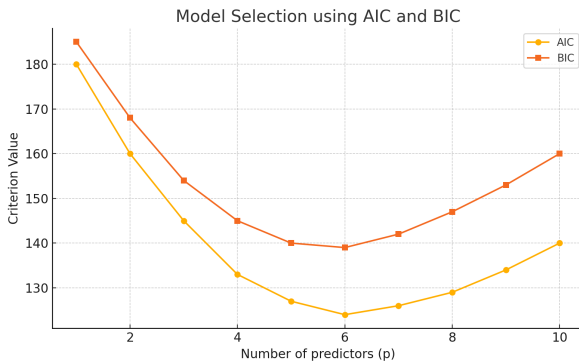
- Lower BIC is better $\Rightarrow$ model is more likely to be true among candidates
- Penalizes complexity more strongly than AIC (because of $\log n$ term)
- As $n \rightarrow \infty$, BIC tends to select the correct (true) model if it exists

Use BIC when:

- You prioritize simpler models and model identification over pure predictive performance
- You are comparing models fitted to the same data

Visualizing AIC and BIC

- Both AIC and BIC decrease as model fit improves.
- BIC typically reaches its minimum earlier than AIC due to a stronger penalty on model complexity.
- Choose the model at the minimum value of each criterion.



Mallows' C_p Criterion

Formula:

$$C_p = \frac{RSS_p}{\hat{\sigma}^2} - n + 2p$$

- RSS_p : Residual Sum of Squares from a model with p predictors
- $\hat{\sigma}^2$: Estimate of the error variance from the full model (with all predictors)
- n : Number of observations
- p : Number of predictors in the subset model (including intercept)

Interpretation:

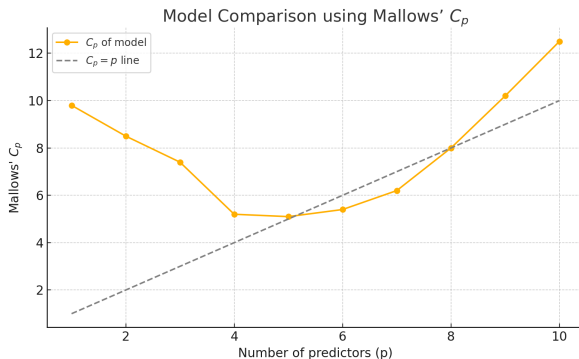
- If the model is **unbiased**, then $C_p \approx p$
- Models with $C_p \approx p$ and **small overall value** are preferred
- Penalizes both lack of fit and model complexity

Use Mallows' C_p when:

- You trust the full model as a benchmark to estimate $\hat{\sigma}^2$
- You want a simple diagnostic to compare subset regression models

Visualizing Mallows' C_p

- Plot C_p values for models with increasing number of predictors
- Look for the model where $C_p \approx p$ and closest to the dashed line
- Avoid overfitting: adding more predictors increases C_p again



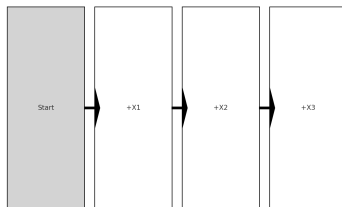
Comparison: AIC vs. BIC vs. C_p

- **AIC:** favors predictive accuracy; more lenient on complexity
- **BIC:** favors simplicity; stronger penalty for extra parameters
- C_p : compares subset models to full model using error-based criterion

Which to use?

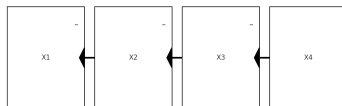
- AIC: when prediction is the primary goal
- BIC: when identifying the “true” model or simpler model is preferred
- C_p : when variance estimate from full model is trustworthy

Forward Stepwise Selection



- Start with no predictors (or intercept only)
- Add the variable that most improves the model
- Repeat until no significant improvement

Backward Stepwise Selection



- Start with the full model
- Remove the variable whose exclusion improves performance most
- Continue until further removal worsens the model

Stepwise Regression

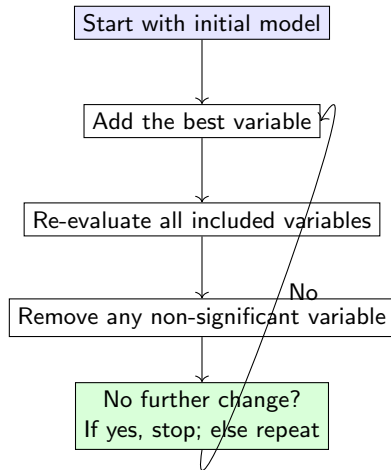
Stepwise regression is an iterative model selection method that combines forward selection and backward elimination.

Procedure:

- 1 Start with an initial model (e.g., intercept-only).
- 2 **Add** the predictor that most improves the model (e.g., lowest AIC or highest F-statistic).
- 3 After adding, **recheck all included variables**.
- 4 **Remove** any variable that no longer contributes significantly.
- 5 Repeat steps 2–4 until no further improvement is possible.

Key point: Variables already added may still be removed later.

Stepwise Selection Flowchart



Stepwise vs. All-Subsets Selection

- Stepwise methods only explore a subset of models
- They are computationally efficient but may miss best models
- **All-subsets selection** considers all 2^p models (feasible only for small p)
- Stepwise methods are a compromise: faster but potentially suboptimal

Variable Selection in R